

Objective

To understand and implement the basic programming constructs of Java, including variables, data types, operators, conditional statements, loops, and methods.

Concepts Covered

- Java Program Structure
- Variables and Data Types
- Operators
- Input and Output
- Conditional Statements (if, if-else, switch)
- Looping Statements (for, while, do-while)
- Methods (User-defined Functions)

Task Description

Develop a "**Student Utility Program**" that performs the following operations using a menu-driven approach. Part A: Java Basics

- Display student information.
- Accept Name, Roll Number, and Marks.
- Calculate Total and Percentage.
- Display the result.

Part B: Conditional Statements

Implement menu options to:

- Check whether a number is Even or Odd.
- Find the Largest of Three Numbers.
- Display Grade based on percentage using if-else.
- Display Day of the Week using switch.

Part C: Looping Statements

Write programs to:

- Print multiplication tables.
- Display numbers from 1 to N.
- Calculate the sum of first N natural numbers.
- Generate the Fibonacci series.

Part D: Methods : Every method should be parameterized and should return the result:

Create user-defined methods for:

- Finding the factorial of a number.
- Checking whether a number is prime.
- Finding the maximum of two numbers.
- Calculating the area of a circle.

Call each method from the main() method.

Algorithm: StudentUtility Program

Step 1: Start the program.

Step 2: Create a `Scanner` object to read input from the user.

Step 3: Display the main menu:

1. Student Information
2. Conditional Statements
3. Looping Statements
4. Utility Methods
5. Exit

Step 4: Read the user's choice.

Step 5: Use a `switch` statement to perform the selected operation.

Step 6: If the user selects **Student Information**:

- Read student name, roll number and marks of three subjects.
- Calculate total marks.
- Calculate percentage.
- Display the student result.

Step 7: If the user selects **Conditional Statements**:

- Display the conditional menu.
- Perform Even/Odd, Largest of Three, Grade calculation, or Day of Week using conditional statements.

Step 8: If the user selects **Looping Statements**:

- Display the looping menu.
- Perform multiplication table, numbers from 1 to N, sum of N natural numbers, or Fibonacci series using loops.

Step 9: If the user selects **Utility Methods**:

- Display the methods menu.
- Call the appropriate method for factorial, prime checking, maximum of two numbers, or area of a circle.

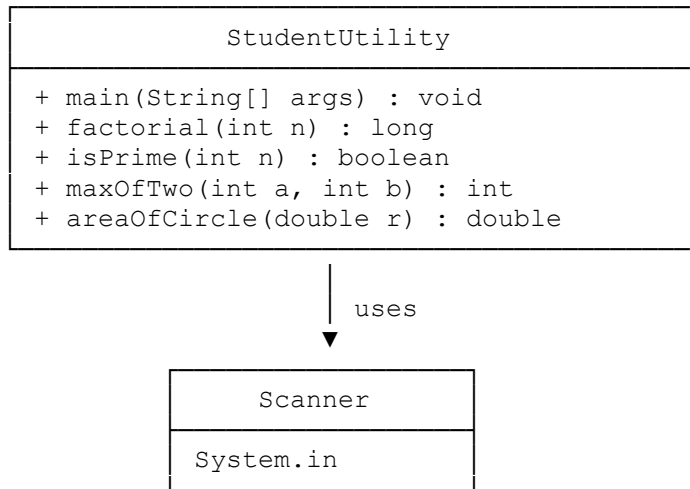
Step 10: If the user selects **Exit**, display the exit message.

Step 11: Repeat the main menu until the user selects option 5.

Step 12: Close the `Scanner`.

Step 13: Stop the program.

UML Class Diagram



```
main()
├── Student Information
│   ├── Name
│   ├── Roll Number
│   ├── Marks
│   ├── Total
│   └── Percentage
├── Conditional Statements
│   ├── Even / Odd
│   ├── Largest of Three
│   ├── Grade
│   └── Day of Week
├── Looping Statements
│   ├── Multiplication Table
│   ├── Numbers 1 to N
│   ├── Sum of N Natural Numbers
│   └── Fibonacci Series
└── Utility Methods
    ├── factorial()
    ├── isPrime()
    ├── maxOfTwo()
    └── areaOfCircle()
```

Code:

```
package javacore;

import java.util.Scanner;

public class StudentUtility {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;
        do {
            System.out.println("StudentUtility program");
            System.out.println("1.Student Information");
            System.out.println("2.Conditional Statements");
            System.out.println("3.Looping Statements");
            System.out.println("4.Utility methods");
            System.out.println("5.Exit");
            System.out.println("Enter your choice:");
            choice=sc.nextInt();
            switch(choice) {
                case 1:
                    System.out.println("Enter Name:");
                    sc.nextLine();
                    String name=sc.nextLine();
                    System.out.println("Enter the Roll Number:");
                    int roll =sc.nextInt();
                    System.out.println("Enter Marks in 3
subjects");
                    int
m1=sc.nextInt(),m2=sc.nextInt(),m3=sc.nextInt();
                    int total=m1+m2+m3;
                    double percentage =(total/3)*100;
                    System.out.println("\n--- Student Result ---");
                    System.out.println("Name: " + name);
                    System.out.println("Roll No: " + roll);
                    System.out.println("Total Marks: " + total);
```

```

        System.out.println("Percentage: " + percentage
+ "%");
        break;

    case 2:
        System.out.println("\n Conditional Statement");
        System.out.println("1. Even or Odd");
        System.out.println("2. Largest of Three
Numbers");
        System.out.println("3. Grade based on
Percentage");
        System.out.println("4. Day of the Week");
        int condChoice = sc.nextInt();

        switch (condChoice) {
            case 1:
                System.out.print("Enter a number: ");
                int num = sc.nextInt();
                System.out.println(num % 2 == 0 ?
"Even" : "Odd");
                break;
            case 2:
                System.out.print("Enter three numbers:
");
                int a = sc.nextInt(), b =
sc.nextInt(), c = sc.nextInt();
                int largest = (a > b) ? (a > c ? a :
c) : (b > c ? b : c);
                System.out.println("Largest: " +
largest);
                break;
            case 3:
                System.out.print("Enter percentage:
");
                double per = sc.nextDouble();

```

```

        if (per >= 90)
System.out.println("Grade: A+");
        else if (per >= 75)
System.out.println("Grade: A");
        else if (per >= 60)
System.out.println("Grade: B");
        else if (per >= 40)
System.out.println("Grade: C");
        else System.out.println("Grade:
Fail");

        break;
    case 4:
        System.out.print("Enter day number (1-
7): ");

        int day = sc.nextInt();
        switch (day) {
            case 1:
System.out.println("Monday"); break;
            case 2:
System.out.println("Tuesday"); break;
            case 3:
System.out.println("Wednesday"); break;
            case 4:
System.out.println("Thursday"); break;
            case 5:
System.out.println("Friday"); break;
            case 6:
System.out.println("Saturday"); break;
            case 7:
System.out.println("Sunday"); break;
            default:
System.out.println("Invalid day!");
        }
        break;
    }
    break;
}

```



```

        int terms = sc.nextInt();
        int f1 = 0, f2 = 1;
        System.out.print("Fibonacci: " + f1 + "
" + f2);

        for (int i = 3; i <= terms; i++) {
            int next = f1 + f2;
            System.out.print(" " + next);
            f1 = f2;
            f2 = next;
        }
        System.out.println();
        break;
    }
    break;
case 4: // Part D
    System.out.println("\n--- Methods ---");
    System.out.println("1. Factorial");
    System.out.println("2. Prime Check");
    System.out.println("3. Max of Two Numbers");
    System.out.println("4. Area of Circle");
    int methodChoice = sc.nextInt();

    switch (methodChoice) {
        case 1:

            System.out.println("Enter number: ");
            int factNum = sc.nextInt();
            System.out.println("Factorial = " +
factorial(factNum));
            break;
        case 2:

            System.out.println("Enter number: ");
            int primeNum = sc.nextInt();

```



```

        System.out.println(isPrime(primeNum) ?
"Prime" : "Not Prime");
        break;
    case 3:
        System.out.print("Enter two numbers: ");
        int x = sc.nextInt(), y = sc.nextInt();
        System.out.println("Max = " + maxOfTwo(x,
y));

        break;
    case 4:
        System.out.print("Enter radius: ");
        double r = sc.nextDouble();
        System.out.println("Area = " +
areaOfCircle(r));
        break;
    }

        break;

    case 5:
        System.out.println("Exiting program...");
        break;

    default:
        System.out.println("Invalid choice!");
    }
} while (choice != 5);

sc.close();
}

    public static long factorial(int n) {
        long fact = 1;
        for (int i = 1; i <= n; i++) {
            fact *= i;
        }
        return fact;
    }
}

```

```
public static boolean isPrime(int n) {  
    if (n <= 1)  
        return false;  
  
    for (int i = 2; i <= Math.sqrt(n); i++) {  
        if (n % i == 0)  
            return false;  
    }  
    return true;  
}  
  
public static int maxOfTwo(int a, int b) {  
    return (a > b) ? a : b;  
}  
  
public static double areaOfCircle(double r) {  
    return Math.PI * r * r;  
}  
}
```

Code screenshot:


```
File Edit Selection View Go Run Terminal Help
sahithi_java

Problems Output Debug Console Terminal Ports
Enter N: 456
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70
71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 12
8 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178
179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228
229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 2
79 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 32
9 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379
380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429
430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456
StudentUtility program
1.Student Information
2.Conditional Statements
3.Looping Statements
4.Utility methods
5.Exit
Enter your choice:
3

--- Looping Statements ---
1. Multiplication Table
2. Display numbers 1 to N
3. Sum of first N natural numbers
4. Fibonacci Series
4
Enter terms: 43
Fibonacci: 0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765 10946 17711 28657 46368 75025 121393 196418 317811 514229 832040 1346269 2178309 3524578 5702887 9227465 14930352 2415781
7 39088169 63245986 102334155 165580141 267914296
StudentUtility program
1.Student Information
2.Conditional Statements
3.Looping Statements
4.Utility methods
5.Exit
Enter your choice:
5
Exiting program...
PS C:\Users\Admin\OneDrive\Documents\sahithi_java>
```